Experiment 7: Developing a Simple Full-Stack Web Application using Http Node Server and MYSQL Database

Source Code: [GitHub Repository]

1) Open Visual Studio Code and launch the integrated terminal.
2) Create the project files: index.html and server.js.
3) In MySQL, create a database named **testdb**, create a table, and insert at least two records using SQL commands.
4) Initialize the project and install dependencies: **npm init -y and npm install mysql2.**
5) Run the frontend by launching index.html using Live Server.
6) Start the backend Node.js server using the command: **node server.js**

Index.html

```html
<!DOCTYPE html>
<html>
<head>
  <title>Users</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f6f8;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
    .container {
```

```
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 4px 10px rgba(0,0,0,0.1);
      width: 350px;
      text-align: center;
    }
    h2 { margin-bottom: 20px; }
    button {
      padding: 10px 15px;
      border: none;
      background: #007bff;
      color: white;
      border-radius: 5px;
      cursor: pointer;
      font-size: 14px;
    }
    button:hover { background: #0056b3; }
    ul {
      list-style: none;
      padding: 0;
      margin-top: 20px;
    }
    li {
      background: #f1f3f5;
      margin: 8px 0;
      padding: 10px;
      border-radius: 5px;
      text-align: left;
    }
    .loading { margin-top: 10px; font-size: 14px; color: gray; }
    .error { color: red; margin-top: 10px; }
  </style>
</head>
<body>
<div class="container">
```

```
  <h2>Users List</h2>
  <button onclick="loadUsers()">Load Users</button>
  <div id="status"></div>
  <ul id="list"></ul>
</div>
<script>
async function loadUsers() {
  const status = document.getElementById("status");
  const list = document.getElementById("list");
  status.innerHTML = "Loading...";
  list.innerHTML = "";
  try {
    const res = await fetch("http://localhost:3000/users");
    if (!res.ok) throw new Error("Failed to fetch");
    const data = await res.json();
    status.innerHTML = "";
    if (data.length === 0) {
      status.innerHTML = "No users found";
      return;
    }
    data.forEach(user => {
      const li = document.createElement("li");
      li.innerHTML =
`<strong>${user.name}</strong><br><small>${user.email}</small>`;
      list.appendChild(li);
    });
  } catch (err) {
    status.innerHTML = "Error loading users";
  }
}
</script>
</body>
</html>
```

Server.js

```javascript
const http = require("http");
const mysql = require("mysql2");
const db = mysql.createConnection({
  host: "localhost",
  user: "root",
  password: "db_server_password",
  database: "testdb"
});
const server = http.createServer((req, res) => {
  res.setHeader("Access-Control-Allow-Origin", "*");
  res.setHeader("Access-Control-Allow-Methods", "GET, POST,
OPTIONS");
  res.setHeader("Access-Control-Allow-Headers", "Content-Type");
  if (req.method === "OPTIONS") {
    res.writeHead(200);
    return res.end();
  }
  if (req.method === "GET" && req.url === "/users") {
    db.query("SELECT * FROM users", (err, result) => {
      if (err) {
        res.writeHead(500);
        return res.end("Database error");
      }
      res.writeHead(200, { "Content-Type": "application/json" });
      res.end(JSON.stringify(result));
    });
    return;
  }
  res.writeHead(404);
  res.end("Not Found");
});
```

```
server.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

Database query :

```sql
CREATE DATABASE testdb;
USE testdb;
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100)
);
INSERT INTO users (name, email) VALUES
('user1', 'user1@gmail.com'),
('user2', 'user2@gmail.com');
```